

A SURVEY ON NON-INTRUSIVE ASR REFINEMENT: FROM OUTPUT-LEVEL CORRECTION TO FULL-MODEL DISTILLATION

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ABSTRACT

Automatic Speech Recognition (ASR) has become an integral component of modern technology, powering applications such as voice-activated assistants, transcription services, and accessibility tools. Yet ASR systems continue to struggle with the inherent variability of human speech, such as accents, dialects, and speaking styles, as well as environmental interference, including background noise. Moreover, domain-specific conversations often employ specialized terminology, which can exacerbate transcription errors. These shortcomings not only degrade raw ASR accuracy but also propagate mistakes through subsequent natural language processing pipelines. Because redesigning an ASR model is costly and time-consuming, non-intrusive refinement techniques that leave the model’s architecture unchanged have become increasingly popular. In this survey, we systematically review current non-intrusive refinement approaches and group them into five classes: fusion, re-scoring, correction, distillation, and training adjustment. For each class, we outline the main methods, advantages, drawbacks, and ideal application scenarios. Beyond method classification, this work surveys adaptation techniques aimed at refining ASR in domain-specific contexts, reviews commonly used evaluation datasets along with their construction processes, and proposes a standardized set of metrics to facilitate fair comparisons. Finally, we identify open research gaps and suggest promising directions for future work. By providing this structured overview, we aim to equip researchers and practitioners with a clear foundation for developing more robust, accurate ASR refinement pipelines.

Keywords Automatic Speech Recognition, Refinement, Non-Intrusive, Adaptation, Datasets, Metrics

1 Introduction

Automatic Speech Recognition (ASR) systems face significant challenges, including variability in speech due to accents, dialects, and speaking styles, as well as environmental factors like noisy backgrounds. Additionally, domain-specific conversations often involve specialized terminologies that can lead to transcription errors [1]. These issues degrade the performance and reliability of ASR applications, necessitating techniques to refine and enhance their outputs.

To address these challenges, researchers have explored four primary approaches: designing comprehensive architectures, training or fine-tuning existing models on custom datasets, developing complementary models, and modifying the training strategy to enhance ASR systems. With the advent of neural networks, particularly transformers [2], numerous deep neural network architectures have been proposed to improve ASR reliability [3–6]. However, designing new architectures often demands significant computational resources and large paired speech-text datasets, which may not always be available. Similarly, training or fine-tuning an existing model on an additional small dataset risks overfitting [5], and in many cases, the model may not be accessible for modification or may exceed users’ computational capabilities. These limitations render such approaches impractical in many scenarios.

In response, researchers have developed methods to refine ASR performance without altering the neural architecture of the model or requiring additional paired speech-text data. Typical ASR refinement techniques rely on rule-based

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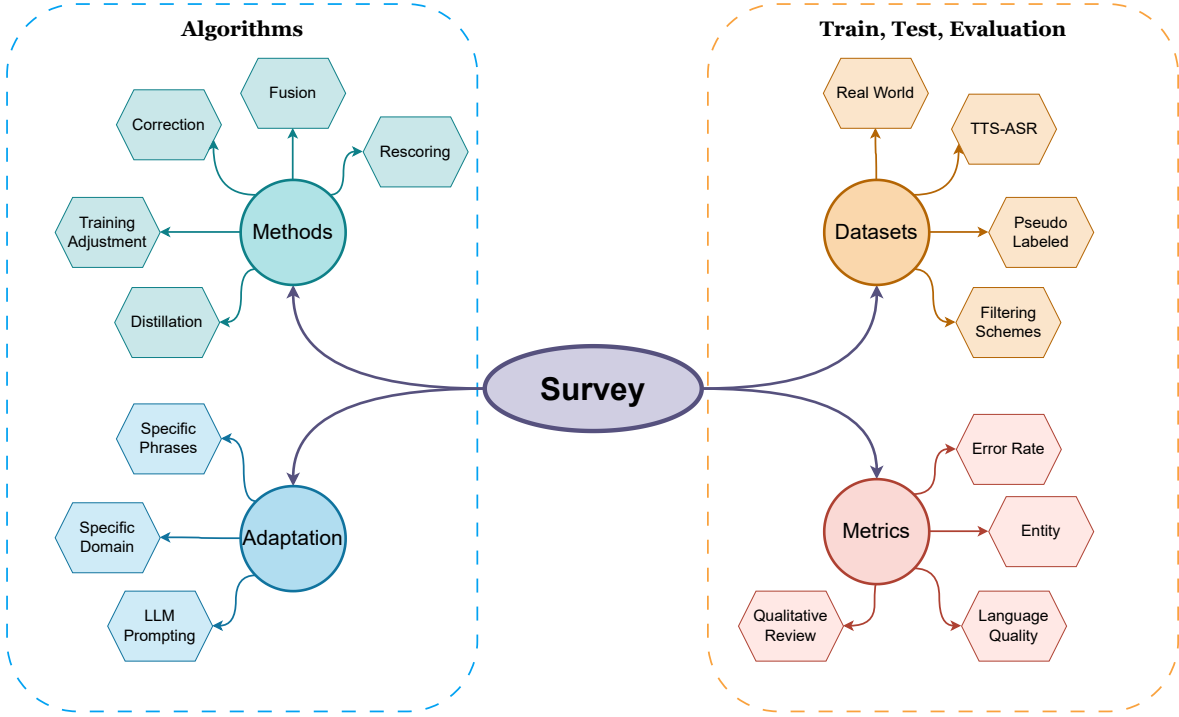


Figure 1: A comprehensive overview of survey sections and subsections.

approaches [7] or n-gram language models [8]. However, these methods do not generalize well because it is impractical to define a comprehensive set of rules capable of addressing the vast linguistic variations. Consequently, their generalizability is severely limited. N-gram methods face similar challenges; moreover, they are constrained by a limited context length (typically up to 4 or 5 words), a critical factor in Natural Language Processing (NLP).

With the advent of more powerful computational processing units and deep neural networks, which excel at learning generalizable features, significant advancements have been achieved in this research area, mirroring progress in many other fields. Researchers began adopting Neural Language Models (NLMs), particularly Recurrent Neural Networks (RNNs), to refine ASR systems [9]. However, despite their ability to handle larger contexts than n-grams, RNNs struggle with long contexts due to their tendency to forget earlier information after a few steps.

The introduction of transformers [2] marked a significant evolution in ASR architectures and their refinement techniques, a trend observed across numerous applications. Many researchers adopted transformers as the backbone of their proposed methods in various forms [10–14], leveraging these models’ ability to process all input tokens simultaneously and selectively focus their attention. However, initial transformer-based methods [15, 16] faced challenges due to their high computational cost, especially during decoding. To mitigate this burden, several solutions were proposed, including encoder-only approaches [12, 17], parallel AutoRegressive (AR) decoding [18], and techniques employing Non-AutoRegressive (NAR) decoding [14, 19].

Recently, the emergence of Large Language Models (LLMs) has influenced ASR and its refinement methods. Various approaches now leverage LLMs for ASR refinement, with recent innovations incorporating Retrieval-Augmented Generation (RAG) to supply relevant context, thereby enhancing the correction capabilities of LLMs.

Additionally, several methods have been developed to enhance ASR performance during its training process or through fine-tuning [20–22]. These techniques do not modify the ASR architecture or require additional data; instead, they improve performance by optimizing the training strategy. These techniques, that we call *Non-Intrusive* in this research, leverage existing training data or external strategies, such as Language Models (LMs) or auxiliary training objectives, to enhance recognition accuracy. Non-intrusive methods are categorized in this survey into five main classes; *Fusion* (Combining outputs from multiple models to improve accuracy), *Rescoring* (Re-ranking hypotheses using language models to select the most likely transcription), *Correction* (Generating improved transcriptions by correcting errors in the initial output), *Distillation* (Transferring knowledge from larger models to smaller ones to enhance performance), and *Training Adjustment* (Optimizing the training process to improve the model’s understanding of existing data).

These methods span a spectrum from output-level corrections, requiring minimal access to the ASR model, to full model distillation, which retrains model parameters using existing resources and needs full access to the model’s parameters.

While prior surveys have addressed specific aspects of ASR refinement—such as human-aided and conventional methods [1], language model integration in encoder-decoder systems [23], LLM-based Chinese error correction [24], and AI-driven medical transcription enhancements [25]—a comprehensive review of non-intrusive techniques remains absent. Moreover, existing literature often overlooks the diversity of dataset creation schemes and evaluation metrics, hindering standardized comparison factors across methods.

This survey addresses these gaps by providing:

1. **Categorization of Non-Intrusive ASR Refinement Techniques:** A detailed classification of techniques into fusion, rescoring, correction, distillation, and training adjustment.
2. **Review of Datasets and Creation Schemes:** An overview of datasets, their creation processes, and filtering methods used in ASR refinement research.
3. **List of Evaluation Metrics:** A compilation of commonly used metrics to facilitate method comparison.
4. **Exploration of Domain-Specific Adaptation Techniques:** An analysis of techniques tailored to specific domains or phrase sets, distinguishing them from general-purpose non-intrusive methods.
5. **Identification of Research Gaps and Future Directions:** A summary of current limitations and proposed research pathways.

The paper is structured as follows: Section 2 details non-intrusive methods, covering fusion, rescoring, correction, distillation, and training adjustment. In Section 3, we discuss domain-specific adaptation techniques, which aim improving ASR transcriptions in a specific domain or on a limited list of phrases. Section 4 reviews datasets used in ASR refinement research, their creation processes, filtering schemes and sources of data for researches in the field. Section 5 introduces the evaluation metrics prevalent in the field, addressing both currently available metrics and the metrics required in the future for a viable comparison. Finally, Section 6 identifies research gaps and proposes future directions. The overall structure of this research is also provided in Figure 1.

2 Methods

In this section, we survey non-intrusive methods for refining ASR systems, classified into five categories:

- **Fusion:** integrating an external LM into decoding,
- **ReScoring:** re-evaluating hypotheses using an LM,
- **Correction:** generating new transcripts from hypotheses,
- **Distillation:** retraining the ASR model with LM knowledge, and
- **Training Adjustment:** enhancing performance through novel training strategies without external LMs.

Figure 2 provides an overview of the main method categories discussed in this section, while Figure 3 offers a detailed breakdown of each category. Additionally, Table 1 presents a side-by-side comparison of these methods, highlighting key practical considerations.

2.1 Fusion

This section reviews methods for integrating external language models (LMs) into end-to-end (E2E) ASR systems. E2E architectures jointly learn acoustic, pronunciation, and language information but cannot exploit large unpaired text corpora without explicit fusion strategies [26]. A broad taxonomy of fusion techniques was proposed in [23], comparing shallow fusion, deep fusion, cold fusion, LM-as-decoder-layer, and LM integration via multi-task training. It is reported that shallow fusion achieves the lowest first-pass word error rate (WER), while cold fusion yields superior oracle WER after second-path rescoring on n-best hypotheses.

This section of the survey focuses exclusively on shallow fusion, deep fusion, and cold fusion methods, reviewing non-intrusive approaches that neither alter the original ASR architecture nor require additional data involvement.

To this end, we first review shallow fusion techniques, followed by deep fusion methods, and conclude with cold fusion.

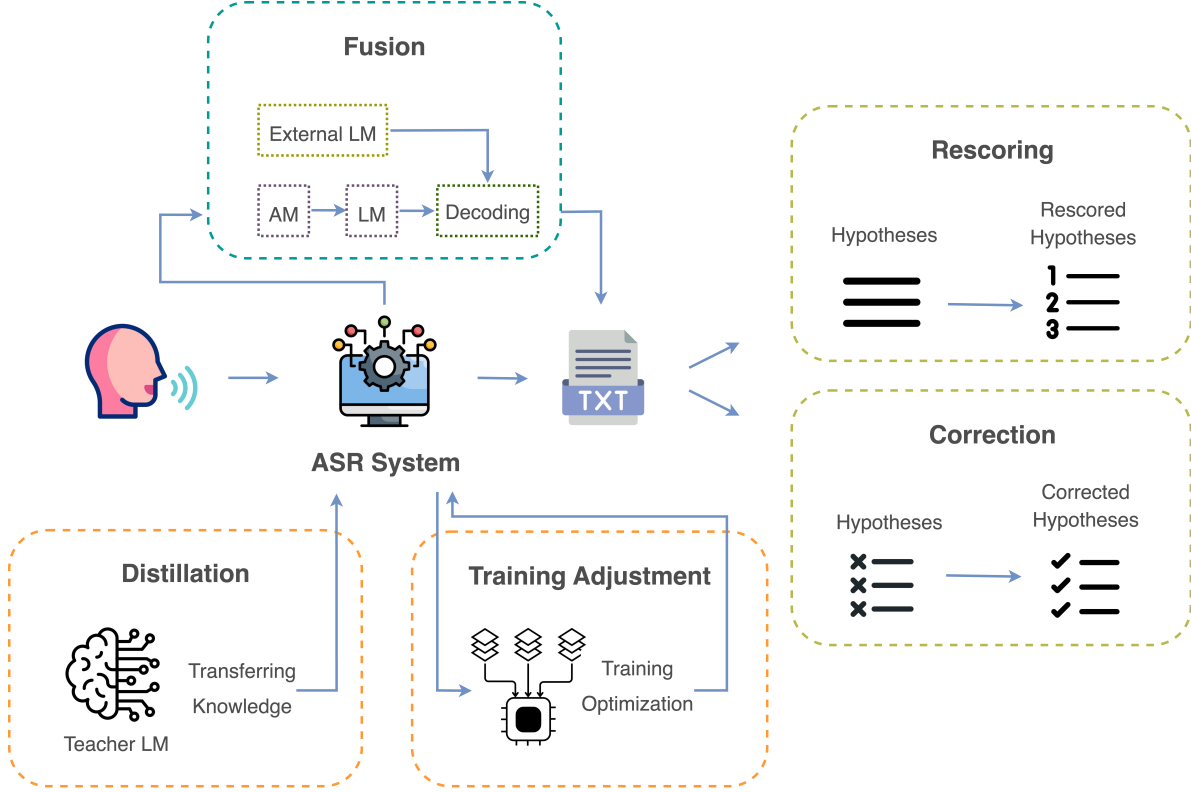


Figure 2: Schematic of ASR refinement methods (AM and LM refer to Acoustic Model and Language Model, respectively).

Shallow Fusion This fusion is an inference-only technique for end-to-end ASR whereby an external LM, trained separately on large text corpora, is integrated during beam-search decoding by log-linear interpolation of the encoder-decoder model’s score and the LM score at each time step [113]. Formally, given an acoustic input x , hypotheses y are chosen to maximize

$$y^* = \arg \max_y [\log p(y | x) + \lambda \log p_{\text{LM}}(y) + \gamma \text{penalty}(x, y)] \quad (1)$$

, where λ and γ are tuned on held-out data. The penalty term is used to control hypothesis length by imposing a fixed cost per decoded token, thereby discouraging overly short or overly long transcriptions and balancing insertion/deletion errors. Several studies have refined shallow-fusion decoding with targeted penalty terms [28–30]. This simple fusion requires no ASR retraining and thus can leverage arbitrarily large unpaired text to improve first-pass WERs [9].

Kannan et al. [9] systematically analyze shallow fusion for sequence-to-sequence ASR by evaluating how different external language model architectures, decoding units, and fusion-weight settings affect inference-time integration. Their study culminates in actionable guidelines for selecting language model types and decoding strategies to maximize shallow fusion gains in practical ASR deployments.

Although shallow fusion effectively incorporates external LMs at inference, it is fundamentally a heuristic interpolation with no clear probabilistic justification and remains biased toward the source-domain internal LM prior [22, 31–33]. To correct this bias, McDermott et al. [32] introduce the *Density Ratio Method* (DRM), which applies Bayes’ rule to subtract the source-domain score from the E2E posterior before adding the target-domain LM score, thereby replacing the source prior with the desired external prior (an external LM is trained on the source domain). The DRM objective is:

$$y^* = \arg \max_y \left[\log p(y | x) + \lambda_T \log p_{\text{LM}}^{\text{target}}(y) - \lambda_S \log p_{\text{LM}}^{\text{source}}(y) \right], \quad (2)$$

where $p_{\text{LM}}^{\text{target}}$ and $p_{\text{LM}}^{\text{source}}$ denote the external and source language model probabilities, respectively. DRM consistently outperforms shallow fusion in cross-domain evaluations [32].

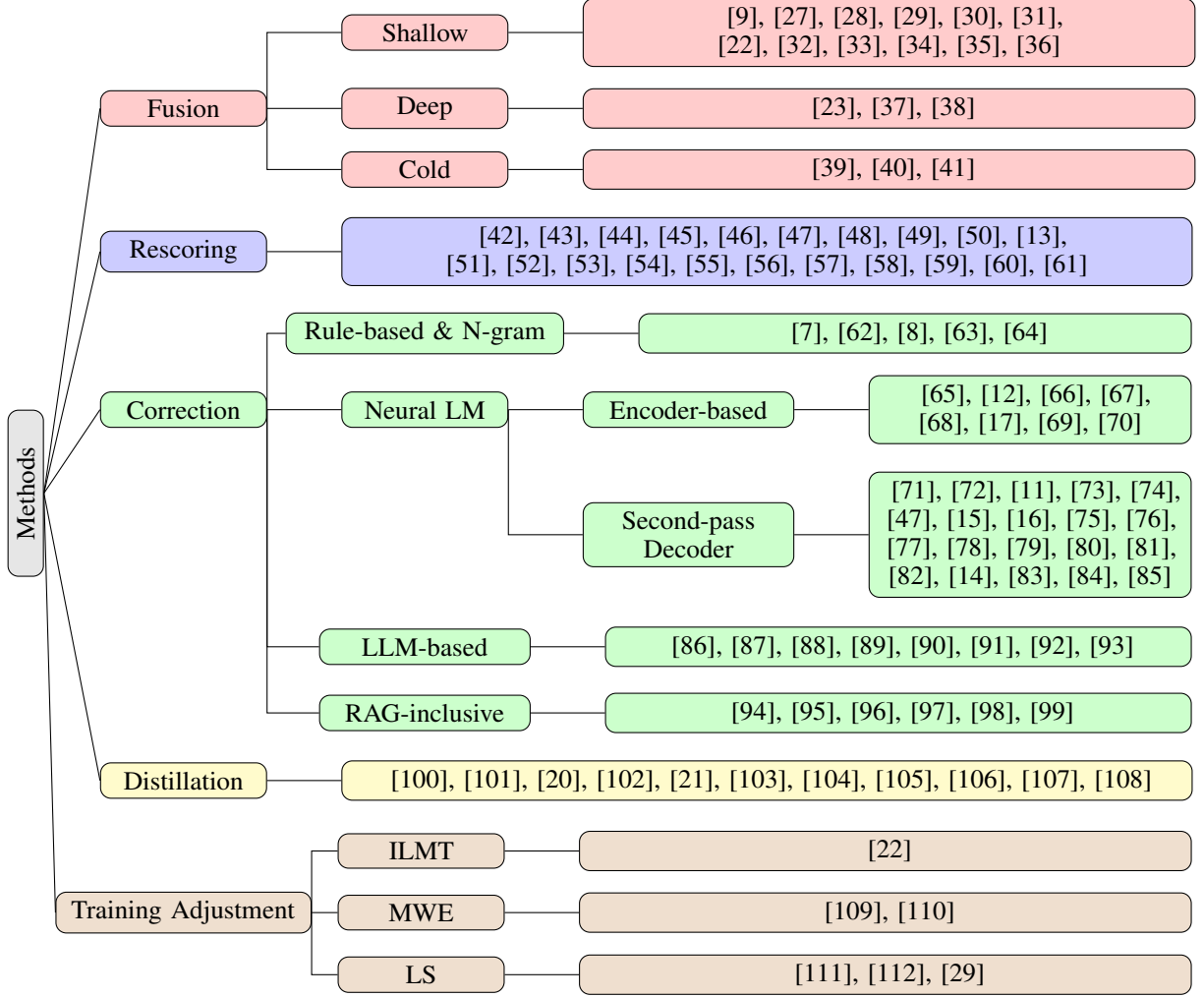


Figure 3: Overview of methods for non-intrusive ASR refinement, grouped into Fusion, Rescoring, Correction, Distillation, and Training Adjustment. Each branch shows subcategories with representative studies.

Compared to *Density Ratio*—which subtracts a separately trained source-domain LM under a hybrid-style factorization—Meng et al. in [31] proposed internal LM estimation (ILME) that employs *Joint Softmax Approximation* to estimate and subtract the E2E model’s own internal LM (ILM) scores during inference, thereby: (1) eliminating the need for an external source LM, (2) correcting the flawed AM–LM decomposition in DRM objective function, and (3) enabling more accurate, efficient fusion.

Ma et al. [33] observe that subtracting a well-trained ILM in standard density-ratio method [32] and ILME [31] can degrade general-domain ASR, especially when the ILM itself is strong; to mitigate this, they propose ILME-ADA, which at inference computes

$$\log P_{\text{ADA}}(y | x) = \log P_{\text{ASR}}(y | x) - \lambda_{\text{ILM}} \log P_{\text{ILM}}(y) + \max\{\lambda \log P_{\text{ELM}}(y), \lambda_{\text{ILM}} \log P_{\text{ILM}}(y)\}, \quad (3)$$

where ELM and ILM stand for external and internal language models, respectively. Thereby, it dynamically selects the more reliable LM score at each decoding step and achieves robust domain adaptation without hurting overall performance.

Ogawa et al. [34] introduce *Iterative Shallow Fusion* (ISF), in which an external *backward language model* (BLM) is applied repeatedly to partial ASR hypotheses during inference, successively replacing previous BLM scores at each decoding iteration. To tailor the BLM to partial sequences, they train a *partial sentence-aware BLM* (PBLM) on

Table 1: Comparison of non-intrusive ASR refinement methods for real-world use at three levels: low, medium, and high.

Aspect	Fusion	Rescoring	Correction	Distillation	Training Adjustment
Trainability	Mixed: shallow (no retrain); deep/cold (trainable)	Inference-only	Trainable (refiner)	Trainable (retrain ASR)	Trainable (loss/regularizer tweaks)
Implementation Complexity	Medium	Medium	Medium-High	High	Medium
Pass Type	First-pass	Second-pass	Second-pass	First-pass	First-pass
Latency	High	Medium	Medium-High	Low (zero)	Low (zero)
Inference Memory	Medium-High	Medium	Medium-High	Low (zero)	Low (zero)
Robustness	High	Medium-High	Low-Medium	High	Medium

reversed text that includes incomplete sentences, improving its efficacy within ISF. They further show that combining ISF with standard shallow fusion using a *forward LM* (FLM) exploits the complementary strengths of BLM and FLM, and that PBLM-based ISF alone achieves performance on par with FLM-based shallow fusion.

Recently, advances in LLMs have also encouraged their deployment in ASR first-pass decoding. Hori et al. [35] introduce delayed fusion, a shallow-fusion-style technique that delays LLM scoring until beam pruning at word boundaries is complete. This approach reduces the number of hypotheses evaluated and costly LLM inference calls, while enabling on-the-fly re-tokenization to handle ASR-LLM vocabulary mismatches. This deferred integration maintains inference latency on par with standard shallow fusion and surpasses the standard Shallow Fusion in error reduction.

Hsu et al. [36] proposed Generative Fusion Decoding (GFD), a shallow fusion method that integrates LLMs with multimodal text recognition systems (ASR, OCR) by mapping text tokens to byte tokens. This enables real-time error correction during decoding without retraining or feature alignment. GFD leverages LLMs’ contextual understanding to improve recognition accuracy, especially for long-form and instruction-aware tasks. Experiments demonstrate significant performance gains across various benchmarks.

Deep Fusion This method augments an end-to-end ASR decoder with a pretrained external LM by learning a small gating network, while freezing both the ASR encoder-decoder and LM weights during fusion training [113]. At each decoding step t , a scalar gate g_t computes how much of the LM hidden state s_t^{LM} to inject into the ASR decoder state s_t :

$$g_t = \sigma(v^\top s_t^{\text{LM}} + b), \quad (4)$$

where $\sigma(\cdot)$ is the Sigmoid activation and $(v \in \mathbb{R}^d, b \in \mathbb{R})$ are trainable [23, 113]. The fused decoder state is then

$$s_t^{\text{DF}} = [s_t; g_t s_t^{\text{LM}}], \quad (5)$$

with $[\cdot; \cdot]$ denoting vector concatenation. Finally, the output hypothesis is

$$y_t = \text{softmax}(\text{DNN}_\theta(s_t^{\text{DF}})), \quad (6)$$

where DNN_θ is a small feed-forward network whose weights θ are also adapted [23, 113].

Recently, Mittal et al. [37] introduced a novel deep fusion method that tightly integrates a pre-trained ASR model with a LLM to enhance transcription accuracy, especially for low-resource languages. SALSA (Speedy ASR-LLM Synchronous Aggregation) synchronously couples the decoder layers of the ASR model (Whisper) with those of the LLM (LLaMA2 7B and 13B) through lightweight projection layers. To address the challenge of differing tokenizations between the ASR and LLM, the method employs cascading tokenization, ensuring seamless interaction between the models.

Generally, while deep fusion initially introduced a principled method for integrating a pre-trained language model with an ASR decoder via a gating mechanism, it has seen limited methodological extensions in recent years. Most subsequent works have shifted toward cold or shallow fusion paradigms, which offer greater flexibility during training or decoding, respectively. Nevertheless, deep fusion retains relevance as a foundational method for evaluating language model integration techniques, particularly in specialized contexts like low-resource language scenarios [38], ensuring its continued role in comparative analyses within the evolving landscape of ASR research.

Cold Fusion Sriram et al. [39] proposed cold fusion that trains a Seq2Seq decoder alongside a fixed pre-trained LM by: (1) projecting the LM’s logits through a DNN to generate features h_t^{LM} , (2) computing a fine-grained gate g_t , (3) fusing the decoder’s hidden state s_t with the gated LM features ($g_t \odot h_t^{\text{LM}}$) into a fused state, and (4) processing this

fused state through another DNN before softmax prediction. Unlike methods that blend LM and decoder scores only during inference or perform partial retraining on gating mechanism, cold fusion requires full retraining of the ASR decoder to deeply integrate the LM’s linguistic knowledge into acoustic modeling.

Cho et al. [40] introduce three memory-control fusion schemes that integrate a frozen RNN-LM into encoder–decoder ASR: the first updates the decoder’s LSTM cell state at each step by adding a gated LM output to the original cell memory; the second combines this cell-control update with the standard cold fusion method [39] by also gating and concatenating LM information into the hidden state before prediction; and the third extends these dual updates with an affine gating mechanism that fuses LM features into both cell and hidden states prior to the softmax layer.

In [41], Kim et al. extend prior cold fusion strategy by enabling an external neural LM to be tightly integrated into an RNN-T both during training and inference without adding any extra algorithmic latency. They introduce lightweight gating and fusion modules that project LM logits into a fusion space and merge them with the transducer’s prediction network in a plug-and-play fashion, allowing any pre-trained Neural Network LM to be swapped in without re-training the core ASR model.

2.2 Rescoring

Rescoring models can range from traditional n -gram language models to more advanced neural language models and large language models. In this method an n -best list is generated with the ASR model, and each hypothesis in the list is rescored. The scoring function can be defined as:

$$\text{Score}(\mathbf{X}, \mathbf{y}) = \log p_{\text{ASR}}(\mathbf{y}|\mathbf{X}) + \alpha \text{Score}_{\text{LM}}(\mathbf{y}) + \beta |\mathbf{y}| \quad (7)$$

where $\mathbf{X}$ denotes acoustic features, $\mathbf{y} = (y_1, \dots, y_L)$ is a hypothesis, and α, β are hyperparameters. $\log p_{\text{ASR}}$ and Score tend to favor shorter hypotheses, so $\beta |\mathbf{y}|$ encourages longer ones to mitigate deletion errors. The Score function can be calculated in different ways, one of which could be for n -gram language models as:

$$\text{Score}_{n\text{-gram}}(\mathbf{y}) = \sum_{i=1}^L \log p(y_i | y_{i-n+1}, \dots, y_{i-1}) \quad (8)$$

where conditional probability p is estimated from training data using maximum likelihood estimation. Chelba et al. [42] discuss large-scale acoustic modeling combined with back-off n -gram language models for Google Voice Search, by leveraging very large n -gram rescoring models. Another work presents lattice rescoring using n -gram language models to improve word accuracy in real-time ASR systems, employing tools like SRI Language Model (SRILM) to assign n -gram probabilities dynamically during decoding.

Rescoring based on neural language model scoring methods can be broadly categorized based on whether they use AR or NAR language models. AR rescoring calculates the score of a hypothesis by sequentially predicting each token conditioned on all previous tokens, also since LLMs are fundamentally AR, they generate text by predicting each token sequentially based on all previously generated tokens and parameter as defined below [43]:

$$\text{Score}_{\text{AR}}(\mathbf{y}) = \sum_{i=1}^L \log p(y_i | \mathbf{y}_{<i}) \quad (9)$$

$$\text{Score}_{\text{LLM}}(\mathbf{y}) = \sum_{i=1}^L \log p(y_i | \mathbf{y}_{<i}; \theta) \quad (10)$$

where $\mathbf{y}_{<i} = (y_1, \dots, y_{i-1})$. and θ denotes the model parameters.

It is to be noted that NAR rescoring evaluates tokens independently or in parallel by conditioning each token on the entire sequence except itself, expressed as:

$$\text{Score}_{\text{NAR}}(\mathbf{y}) = \sum_{i=1}^L \log p(y_i | \mathbf{y}_{\setminus i}), \quad (11)$$

where $\mathbf{y}_{\setminus i} = (y_1, \dots, y_{i-1}, [\text{MASK}], y_{i+1}, \dots, y_L)$ and it enables bidirectional context and faster computation. Next, we examine the historical trend of articles on the use of various rescoring methods mentioned earlier.

Chen et al. [44] employed RNN-based language models to refine transcript-level hypotheses. Their first approach clusters history contexts in an n -gram style to enhance contextual modeling, while their second method leverages the distance between recurrent hidden history vectors to align hypotheses with more probable linguistic patterns. A

different direction is taken by Kumar et al. [45], who utilize LSTM language models for lattice rescoring. Their work explores n -gram style clustering of history contexts and applies a distance-based method for hypothesis selection.

An attention-based discriminative language model for rescoring, trained with a minimum word error rate (MWER) criterion, is presented by Gandhe et al. [46]. Their model attends to both word-level context and acoustic embeddings and incorporates lattice weights to enhance hypothesis selection accuracy, focusing on rescoring rather than generating new hypotheses. A two-pass architecture has been introduced by Hu et al. [47] that involves a rescoring method as a core part of its approach. The first pass generates streamed hypotheses, and the second pass uses a deliberation network to rescore these hypotheses. The deliberation model attends to both the acoustic input and the first-pass hypotheses simultaneously.

Lattice-attention mechanisms within encoder-decoder frameworks have been shown to improve rescoring accuracy, as described by Pandey et al. [48]. Their model encodes lattices with a recurrent network and applies attention to both lattices and acoustic features. A domain-aware second-pass rescoring framework was proposed by Liu et al. [49]. The method fine-tunes a domain-general NLM on domain-specific data and uses an LSTM-based domain classifier to select the appropriate domain-adapted model for rescoring the ASR outputs.

Further efficiency in error correction is achieved by replacing LSTM layers with Transformer layers in the second-pass rescorer, as demonstrated by Li et al. [50]. Their architecture processes entire hypothesis sequences in parallel and interleaves self-attention and cross-attention layers to integrate acoustic and linguistic contexts. Hu et al. [13] enhanced a deliberation model which replaces LSTM layers with transformer layers for second-pass rescoring. Their transformer-based deliberation network uses generalized encoder-decoder attention for parallel processing of audio frames and first-pass hypotheses. Another transformer-based model has been proposed for second-pass rescoring with discriminative training objectives to optimize hypothesis ranking by Udagawa et al. [51] integrate contextual information from large models to refine n -best lists.

A learning-to-rescore mechanism that combines textual features such as BERT sentence embeddings, topic vectors, and perplexity scores from different LMs is introduced by Song et al. [52], while Shin et al. [53] adapt the BERT model to rescore n -best ASR hypotheses without fine-tuning, leveraging masked language modeling within sentences to bridge the gap between training and inference. Another BERT-based masked language model is used by Salazar et al. [54] to compute pseudo-log-likelihood (PLL) scores for rescoring.

Also, Xu et al. [55] fine-tuned a BERT-based model with MWER loss for n -best rescoring, combining masked language modeling pretraining with discriminative objectives and knowledge distillation. Chiu et al. applies BERT-based language models to rerank ASR n -best hypotheses, which are fine-tuned on ASR outputs and their contexts to capture semantic and syntactic correctness beyond traditional n -gram or LSTM LMs [56]. By incorporating deep contextual embeddings, the reranker improves selection of the best hypothesis. Discriminative objectives such as MWER are also used by Shivakumar et al. [57] to fine-tune pretrained causal and bidirectional language models. Parameter-efficient adaptation is addressed by Yu et al. [58], who propose a Low-Rank Adaptation (LoRA) method for adapting pretrained BERT models for ASR rescoring with minimal updates.

Another paper proposed three novel approaches to incorporate personalized content into BERT-based discriminative speech recognition rescoring: gazetteers, prompting, and cross-attention based encoder-decoder models. The methodology builds on the RescoreBERT architecture, which uses a pre-trained BERT model with a feed-forward network attached to the classification token embedding to predict sentence-level scores for ASR hypotheses. Kolehmainen et al. [59] employed a discriminative training approach using MWER loss to directly optimize the expected number of word errors.

Tur et al. [60] have combined instruction-tuned LLMs with confidence scores and prompt-based hypothesis generation, introducing zero-shot methods that dynamically expand hypothesis sets and use LLM scoring to select the most accurate transcript. Context augmentation with LLMs and ranking strategies as explored by Asano et al. [61], ranks n -best ASR hypotheses based on their lexical and semantic similarity to dialogue context and ranks context by phonetic correspondence with ASR outputs. It is enhanced by generating task variations using an LLM, enabling flexible handling of dialogue states.

2.3 Correction

In contrast to fusion, which is a first-pass improvement strategy, and rescoring, which refines scores to select a candidate from ASR hypotheses in a second-pass manner, correction approaches actively generate a revised transcript during the second pass. Unlike other ASR refinement strategies that merely select from existing hypotheses or tweak the first-pass output, the correction category can generate an entirely new transcription, often with an expanded vocabulary. Correction models can introduce vocabulary beyond the original ASR hypotheses by using:

- a simple n -gram or rule-based approach (conventional)
- a neural language model (NLM)
- a large language model (LLM)
- retrieval of external information from a custom knowledge base

In this section, we evaluate the above correction methods and highlight the key contributions in each category. Although many methods combine multiple components (e.g., error detection, confidence estimation) to perform targeted operations, we group them according to their primary innovations, which still fall within the categories outlined above.

Furthermore, we go through LLM-based approaches. With the introduction of LLMs, many researchers developed ideas to explore the use of these models in ASR refinement, leveraging their exceptional generalizability. Finally, we introduce the branch of research that employs RAG in their correction pipeline. Using RAG has also been explored to incorporate information from a custom knowledge base into the correction pipeline. In the following subsections, we have introduced the methods presented in each category, highlighting their contributions and the purposes of the proposed methods.

2.3.1 Rule-based and N-gram

Early ASR correction efforts relied on rule-based algorithms and n -gram methods to refine transcripts. However, both approaches exhibit significant limitations. Rule-based systems struggle with unseen error patterns, since ASR mistakes vary widely across acoustic domains. N -gram models are constrained by their fixed context window: no finite n -gram can cover every valid word or character combination, and choosing a large n incurs high memory and computation costs, and can even introduce over-corrections, while a small n restricts context awareness. Below, we summarize key contributions in this category.

In an early effort for ASR correction, [7] proposed the Recognizer Output Voting Error Reduction (ROVER) method, which first aligns 1-best hypotheses from multiple ASR systems into a Word Transition Network (WTN), and then adopts a voting scheme to select the best token for each position in the transcript. Recently, [62] extended ROVER by using N -best hypotheses from a single ASR system instead of 1-best hypotheses from N different systems, broadening ROVER’s applicability when multiple ASR models are unavailable. Bassil et al. in [8] developed a three-stage pipeline based on n -gram. It first flags errors in any word absent from a unigram vocabulary, then generates correction candidates using character-level bigrams (i.e., letter pairs), and finally selects replacements based on 5-gram frequency statistics over the four preceding words plus the candidate. Anantaram et al. [63] introduced a four-stage pipeline that slides an n -gram window over the transcript to match spans against a lexicon via (a) exact matches, (b) phonetic similarity (sound-alike words), or (c) Levenshtein distance. Unmatched words are then re-evaluated in context for better alternatives, an LSTM-based sequence-to-sequence model is applied to the remaining segments, and a final rule-based pass uses POS patterns or external grammar tools. Zietkiewicz et al. [64] offered a lightweight rule-based scheme that tags tokens with edit operations (append, delete, join) derived from the Ratcliff–Obershelp similarity algorithm. During testing, a neural network decides which edits to apply, and a rule-based module executes them—allowing for efficient, targeted corrections where needed.

2.3.2 Neural LM

With the advent of neural networks—particularly after the introduction of transformers [2]—and advances in computational hardware, there has been a surge of methods aimed at improving ASR post-editing, in addition to the advances in ASR modeling itself. These neural approaches benefit from their ability to capture much longer dependencies than n -gram LMs—a pivotal advantage for many NLP tasks—and from their greater generalization capacity.

We divide neural LM-based correction methods into two categories: encoder-based and decoder-inclusive (second-pass decoding). Encoder-based methods primarily perform context-aware spelling or token corrections over the ASR output. Decoder-inclusive methods go one step further: they re-decode (or “second-pass decode”) using textual, acoustic, or combined information to extend and correct the original ASR hypotheses. Below, we summarize key contributions in each category to clarify their distinct scopes.

Encoder-based Shifting to NLMs, a distinct branch of methods leverages the exceptional contextual understanding of encoder architectures—particularly transformer-based BERT [65]—to drive ASR correction. Although encoder models also play vital roles in hybrid approaches, here we focus exclusively on those that exploit only the encoding mechanism, offering a lightweight yet effective solution for ASR post-editing.

A common baseline in this paradigm is a pretrained (or further trainable) encoder-based transformer (mainly BERT [65]) model tasked with predicting correction candidates at each token position [12, 66]. Because BERT is trained with a masked language modeling (MLM) objective—analogueous to predicting missing or erroneous tokens—it naturally

lends itself to error correction, as demonstrated in [67]. However, the vanilla BERT models evaluated in [12, 66] often struggle to improve transcription accuracy, primarily due to a domain mismatch between their pretraining data and the target test domains. To address these challenges, various adaptation and enhancement methods have been proposed, which we detail below.

Because contextual encoders like BERT predict embeddings based solely on context, they can sometimes replace an ASR token with a semantically plausible but incorrect word, degrading overall correction quality. To avoid this, it’s important not only to consider contextual fit but also symbolic (character-level) and phonological similarities. To this end, Hong et al. [68] introduced FASpell, which first generates correction candidates via BERT’s contextual representations, then uses a Confidence-Similarity Decoder (CSD) to score each candidate by phonological similarity (how words sound), visual similarity (how they look), and semantic coherence within the sentence. Cheng et al. [17] proposed SpellGCN, which employs a graph neural network to encode relationships among words based on pronunciation and visual likeness, and then merges those graph-based features with BERT’s semantic embeddings to improve token-level corrections. Zhang et al. [66] presented MLM-Phonetics, which softly masks tokens based on their phonetic embeddings—biasing BERT toward replacements that preserve pronunciation. Fan et al. [69] built a BERT framework that fuses word-level and phonetic features, using dynamic error-scaling to modulate correction strength. To prevent overcorrection, several pipelines introduce an explicit error-detection module. Zhang et al. [12] advanced this further with Soft-Masked BERT, coupling a bi-GRU error detector with a soft-masking mechanism that interpolates between the original and [MASK] embeddings, according to detection confidence before feeding them into the correction module. Chen et al. [70] similarly integrate an error-detection classifier and a post-correction step driven by phonetic similarity.

Second-pass Decoding Second-pass decoding methods constitute another paradigm within NLM-based error correction. These approaches may employ either encoder-decoder or decoder-only architectures. By incorporating an explicit decoding step, they extend beyond encoder-only methods to correct a broader range of ASR output errors. However, in addition to the challenges discussed in Section 2.3.2, this strategy introduces its own trade-offs: being generative, these methods often incur a higher risk of over-correction compared to encoder-only models, and the AR decoding step adds both computational cost and inference latency. To mitigate these drawbacks, numerous studies have proposed multi-stage pipelines and NAR decoding techniques. Below, we review these efforts.

In a baseline method, Tanaka et al. [71] introduced Neural Error Correction Language Models (NECLMs) with an encoder-decoder RNN architecture that explicitly learns to correct ASR errors. The model is trained on pairs of erroneous and corrected transcripts, learning to predict the corrected sequence given the noisy ASR output.

In a similar branch of studies, Hrinchuk et al. [72] proposed a simple encoder-decoder and transformer-based model to translate ASR outputs into corrected text. Zhang et al. [11] also applied an encoder-decoder Transformer to Connectionist Temporal Classification (CTC) based ASR, augmenting its training step with samples from N-best hypotheses. Gu et al. [73] proposed a denoising language model (DLM) trained with a vast amount of synthetic noisy data, improving its performance using augmentations. Adnew et al. [74] also applied a similar method to a low-resource ASR system. However, the reliance on text modality alone limits the effectiveness of these approaches. Addressing this, Hu et al. [47] adopted the deliberation network concept from Xia et al. [15], introducing a two-pass model with a shared encoder and dual decoders to integrate speech embeddings and text hypotheses via a Listen, Attend, and Spell (LAS) decoder.

Building on this approach, Tanaka et al. [16] employed a transformer-based model with cross-modal encoding, jointly processing textual and acoustic embeddings before decoding—unlike prior methods that encoded modalities separately, such as in Hu et al. [47]. When paired with shallow fusion, their approach excels at capturing cross-modal dependencies, enhancing the correction performance. Li et al. [75] also suggested another encoder-decoder correction method that directly extracts speech embeddings using HuBERT [76] and, after applying cross-modal attention between these embeddings and text embeddings, decodes the corrected transcript. To handle the computational cost and latency caused by AR structures in a similar approach, Wang et al. [77] introduced a NAR align-refine decoding method for streaming RNN-T, aligning and refining outputs efficiently. Moreover, Hu et al. [78] scaled deliberation to multilingual ASR, handling diverse languages. Besides, Kumar et al. in [79] adopted visual information to guide the correction process through two different schemes: a fusion-based algorithm which encodes text and image separately and fuses them for decoding, and a prompt-based procedure that extracts a caption for the corresponding image and concatenates it with the transcription for feeding into an encoder-decoder correction model.

In another essential method, researchers have utilized N-best hypotheses or attention mechanisms to capture more context in their correction methodologies and better guide the process. Ma et al. fine-tuned a pretrained T5 model using ASR N-best hypothesis lists as input, concatenated with special tokens to separate hypotheses and a prefix "text correction:" to guide the model, benefiting from the multi-task pretraining of T5 and the context provided by N-best hypotheses. This approach also employs constrained decoding techniques based on the N-best list to guide its

generation [80]. Guo et al. proposed a hybrid correction model for E2E ASR that generates M correction candidates for each hypothesis in the N -best list using an encoder-decoder architecture and employs a rescoring mechanism combining ASR, LM, and correction scores to select the best transcription [81]. Additionally, Guo et al. added a global attention mechanism for Chinese spelling correction to capture longer context in their correction process [82].

Moreover, targeted modules for confidence estimation, length prediction, and error detection are also adopted in second-pass decoding scenarios to address the challenges of over-correction and latency in second-pass decoding methods. In a comprehensive method, Shu et al. proposed a NAR error correction framework that leverages a pretrained Confidence Estimation Module (CEM) to pinpoint and locate token-level ASR errors [14]. Additionally, N -best hypotheses are aligned via FastCorrect2-style edit paths [83], length-normalized, and then fused by linearly combining word embeddings according to the confidence scores to create joint representations capable of recovering deletions, insertions, and substitutions. Finally, a NAR decoder uses cross-attention over these fused embeddings and the original ASR output to produce corrected transcriptions. Similarly, Du et al. enhanced ASR error correction by integrating a length predictor and a confidence estimation module in a cross-modal framework; the predictor estimates the correct transcription length using acoustic and textual features, while the estimation module identifies unreliable tokens, together improving correction accuracy and enabling utterance rejection [84]. In a distinct method, D’Haro et al. [85] used phrase-based machine translation for correction.

2.3.3 LLM-based

Advancement of LLMs and their strong generalization capabilities has made them useful across many applications, including ASR post-refinement. In ASR error correction, LLMs have been applied in three main scenarios: text-only prompting schemes, the incorporation of auxiliary information in the first manner, and even fine-tuning. In this subsection, we list methods that leverage LLMs as the core component in second-pass ASR error correction. Despite the similar procedure with the second-pass decoding methods in 2.3.2, we devote a separate subsection to generative correction based on LLMs to emphasize its highly-attended research branch. LLMs are also being used in rescoring and fusion methods which are discussed in Sections 2.1 and 2.2, particularly because of their limited exploration in the literature.

Ray et al. [86] proposed a baseline prompting method (zero- or few-shot fine-tuning) for LLMs, aiming to transform noisy ASR outputs into correct text. It leverages paired ASR hypotheses and reference data or N -best lists, occasionally employing constrained decoding to avoid overcorrection. This non-intrusive approach improves transcription accuracy without modifying the ASR system, achieving significant WER reductions across diverse domains and languages.

Adedeji et al. [87] employed Chain-of-Thought prompting in a multistep reasoning process to benefit lower-performing ASRs. Tang et al. [88] advocated full-text corrections—including punctuation and code-switching—rather than partial edits. Fathallah et al. [89] used LLMs to refine video captions, enhancing readability. Pu et al. [90] introduced a multi-stage pipeline: uncertain segments are identified via N -best hypothesis uncertainty estimation, then corrected by an LLM through rule-based, multistep prompts, yielding 10–20 percent relative WER improvements across domains and in zero-shot settings. Li et al. [91] proposed Pinyin-Enhanced Generative Error Correction (PY-GEC), which inputs both the ASR transcript and its Pinyin representation into an LLM to address pronunciation-related errors in Chinese. Trained on synthetic ASR errors with a multitask objective aligning Pinyin and text embeddings, PY-GEC uses only the one-best hypothesis and its Pinyin at inference, outperforming text-only methods on homophone and phonetic-confusion benchmarks. Ghosh et al. [92] introduced a method that refines the top hypothesis by conditioning an LLM on words from N -best lists and on visual features. Udagawa et al. [93] first apply a conservative data-filtering scheme to exclude low-quality ASR hypothesis–reference pairs (see Section 4.4), and then fine-tune a Japanese LLM that ingests both transcript tokens and source phonemes.

2.3.4 RAG-inclusive

Correction methods often incorporate contextual information to improve performance. A recent paradigm—mainly developed for chatbots—involves retrieving customized information to generate more specific responses. Similarly, in ASR error correction, RAG techniques retrieve relevant knowledge based on candidate outputs, bolstering corrections for rare terms. These methods leverage a tailored knowledge base to inform the correction process, mitigating errors in specialized terminology and named entities and learning from past mistakes.

Ghosh et al. [94] introduced DARAG, which augments the training set with synthetic ASR error–correction pairs generated using large language models and text-to-speech systems (see Section 4.2), simulating a wide variety of realistic errors beyond those in the original data. This synthetic data helps the generative error correction (GEC) model handle diverse and unseen errors. DARAG also incorporates retrieval-augmented inputs by fetching relevant named

entities from a database, improving correction of novel or rare entities. This combined data- and retrieval-augmentation strategy enhances the model’s robustness and generalization.

Zhang et al in [95] retrieve named entities from a pre-built vector database—using the ASR transcript as a query—to find closely related candidates. These entities are then provided as context to an LLM that performs error correction on the ASR output. By grounding corrections in external knowledge, the model effectively fixes errors in rare or uncommon entities without altering the ASR system.

Wang et al. [96] proposed DANCER, which corrects named-entity errors by combining phonetic similarity with semantic information from entity descriptions. It detects corrupted entities in the ASR output, retrieves candidate replacements based on phonetic similarity, and uses an entity-description-augmented MLM (EDA-MLM) to rank candidates via transcript context and semantic embeddings. A weighting mechanism balances phonetic and semantic scores, and an entity-rejection step prevents unnecessary corrections, significantly reducing character error rates for phonetically confusing entities.

GEC-RAG [97] corrects ASR errors by first generating transcriptions via a black-box ASR model, then retrieving lexically similar examples from a knowledge base of ASR transcription and ground-truth pairs using TF-IDF and cosine similarity. These examples—showcasing typical error patterns—are fed into an LLM (GPT-4o) through in-context learning prompts, exposing the model to domain-specific errors without modifying the ASR system. Xiao et al. [98] retrieve context by querying a text-based retriever with first-pass ASR hypotheses and custom vocabulary lists, enabling dynamic contextual control. The system then fine-tunes an LLM on the retrieved context, improving recognition of uncommon terms in the retrieved domain.

To avoid failures when ASR output is highly corrupted, Li et al. [99] generate embeddings directly from the utterance audio using a multimodal speech–text embedding network trained to align audio and text embeddings. At inference, the system queries a database of candidate corrections using the audio-derived embedding and retrieves the nearest neighbor. The candidates are then scored by embedding distance to the input audio. Therefore, this method avoids errors from inaccurate ASR hypotheses and achieves higher precision.

2.4 Distillation

Distillation in the context of LMs and ASR training is a technique where a complex, pretrained LM (the teacher) transfers its knowledge to a simpler ASR model (the student) during the training phase. The language model, trained on large-scale text corpora, injects linguistic knowledge into the ASR model to improve its performance, particularly in scenarios with limited paired speech-text data. This process enhances ASR accuracy by leveraging the LM’s deep syntactic and semantic understanding, without increasing computational cost at the inference time.

In [100], the authors integrate a Recurrent Neural Network Language Model (RNNLM), which operates in a left-to-right manner, to enhance a sequence-to-sequence (Seq2Seq) ASR model through distillation. The RNNLM generates soft labels in the form of token probability distributions, which serve as auxiliary supervision for the student ASR model. The loss function combines two components: a hard loss, defined as the cross-entropy between ASR outputs and ground truth transcriptions, and a soft loss that encourages the student to mimic the teacher’s distribution. The combined objective is expressed as:

$$L_{KD} = (1 - \lambda) L_{\text{hard}} + \lambda L_{\text{soft}}, \quad (12)$$

where the soft loss is computed as:

$$L_{\text{soft}} = \sum_i P_i^{\text{teacher}} \log \frac{1}{P_i^{\text{student}}}. \quad (13)$$

Since the teacher’s parameters are fixed during distillation, its output distribution P_i^{teacher} is constant and can be dropped from the numerator of the logarithm in the KL-divergence, yielding this simplified form. Teacher distribution is softened via temperature-scaled softmax:

$$P_i^{\text{teacher}} = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}, \quad (14)$$

where z_i denotes the pre-softmax activation for token i .

Extending beyond this unidirectional framework, [101] adopts a contextual language model, known as the Causal Cloze Completer (COR), which uses self-attention to capture dependencies from both the left and right of each token. This enables the generation of soft labels that incorporate the full context of the input sequence, helping the ASR model to better learn long-range linguistic relationships. Building on this idea, [20] employs BERT as the teacher. In contrast to earlier approaches using unidirectional or shallow bidirectional models, BERT is a deeply bidirectional transformer

trained on large-scale corpora. BERT’s bidirectional nature is deeply integrated, allowing it to provide a more nuanced linguistic understanding, potentially extending context beyond the current utterance to improve ASR accuracy.

Extending BERT distillation to CTC, [102] proposes aligning BERT’s token-level soft distributions with CTC’s frame-level outputs by computing the most probable CTC path via the forward-backward algorithm. Frames aligned to each token are trained to mimic BERT’s softened probabilities (via KL divergence), enabling rich LM knowledge transfer during training while retaining CTC’s fast, greedy decoding and zero runtime cost for BERT. While [102] aligns BERT’s token-level distributions to CTC frame-level outputs using the most probable CTC path, it focuses solely on output-level distillation. In contrast, [21] introduces a more comprehensive framework by employing multiple decoders during training and applying knowledge distillation to both final and intermediate encoder layers. Specifically, it combines attention-based decoders (used only for training) with intermediate CTC losses to inject BERT’s linguistic knowledge into internal representations of the CTC model. This approach improves performance without altering the decoder-free, parallel inference characteristic of CTC models.

Bai et al. in [103] propose a fundamentally different strategy. Instead of distilling token-level probability distributions and minimizing KL divergence, this method transfers knowledge at the level of semantic representations. Specifically, the approach aligns the hidden states of the ASR decoder with those of BERT using Mean Squared Error (MSE) loss. Extending this concept, [104] applies similar embedding-level distillation to attention-based and transducer-based ASR models, enabling knowledge transfer from BERT to models beyond the NAR architecture considered in [103].

Building on these embedding-level strategies, [105] introduces Distill-L2S, a multilingual CTC-based ASR distillation framework that aligns hidden states of a cross-lingual text model (InfoXLM) with those of a speech model (XLSR-wav2vec 2.0). Without changing the CTC decoder, it transfers rich LM knowledge directly into the ASR encoder, yielding consistent gains on low-resource languages.

In [106], the authors use multiple heads in the acoustic model of the hybrid ASR model: one for the hard loss and one for the soft loss, to separate the supervised learning and distillation objectives. This allows the distillation loss to be applied from LMs with different output units, and the heads are trained in a hierarchical multi-task fashion. This approach enables the use of language models with different output units and improves robustness to the two hyperparameters T and λ , due to the separation of the output heads. Similarly, [107] proposes a hierarchical distillation framework that transfers LM knowledge into both the acoustic and linguistic components of an E2E ASR model. To ensure token-level alignment between speech and text, they adopt a CIF-based ASR model [108], which produces one acoustic representation per token by integrating encoder outputs over time. At the acoustic level, contrastive, MSE, and cosine losses are used to align BERT token embeddings with CIF outputs, while at the linguistic level, an MSE loss aligns BERT representations with the ASR decoder’s pre-softmax outputs—similar to what given in [103] and [104].

2.5 Training Adjustment

Within the framework of non-intrusive ASR refinement, where the model architecture remains unchanged and no additional training data is introduced, optimizing the training process itself emerges as a key strategy for enhancing performance. This subsection surveys approaches that adjust the learning process to better target transcription accuracy, incorporate complementary training signals, or regulate model confidence for improved robustness and generalization.

Internal Language Model Training (ILMT) Meng et al. in [22] propose ILMT that augments the standard E2E loss with an *ILM loss* that encourages the acoustically-conditioned decoder (for attention-based encoder decoder model) or the prediction and joint networks (for RNN-T) to behave like a standalone language model. In case of the RNN-T, the ILM loss is defined as:

$$L_{\text{ILM}}(\theta) = - \sum_{u=1}^U \log P(y_u | Y_{<u}; \theta), \quad (15)$$

the combined ILMT loss becomes,

$$L_{\text{ILMT}}(\theta_{\text{RNN-T}}) = L_{\text{RNN-T}}(\theta_{\text{RNN-T}}) + \alpha L_{\text{ILM}}(\theta), \quad (16)$$

where $Y_{<u} = Y_{0:u-1}$, $\theta = (\theta_{\text{pred}}, \theta_{\text{joint}})$, and α weights the ILM term. For AED, the ILM loss is

$$L_{\text{ILM}}(\theta_{\text{dec}}) = - \sum_{Y \in \mathcal{D}} \sum_{u=1}^{U+1} \log P(y_u | Y_{<u}; \theta_{\text{dec}}), \quad (17)$$

and the ILMT loss is

$$L_{\text{ILMT}}(\theta_{\text{AED}}) = L_{\text{AED}}(\theta_{\text{AED}}) + \alpha L_{\text{ILM}}(\theta_{\text{dec}}). \quad (18)$$

By jointly minimizing $L_{\text{E2E}} + \alpha L_{\text{ILM}}$, ILMT updates only the decoder (or prediction and joint) parameters for the LM term, while still training the full model for ASR accuracy.

Minimum Word Error (MWE) Training Hori et al., in [109], propose a training strategy that adjusts the model’s parameters to directly minimize the expected number of word errors by computing the weighted sum of edit distances between the correct transcript and all hypothesized outputs, where the weights are the model’s posterior probabilities. Specifically, for each utterance k , given an N -best list $\{W_{k,n}\}_{n=1}^N$ of candidate transcriptions based on acoustic observations O_k , the MWE risk is defined as:

$$L(\Lambda) = \sum_{k=1}^K \sum_{n=1}^N E\left(W_k^{(R)}, W_{k,n}\right) P_{\Lambda}(W_{k,n} | O_k), \quad (19)$$

where $E\left(W_k^{(R)}, W_{k,n}\right)$ represents the number of word errors (insertions, deletions, substitutions) between the reference sequence $W_k^{(R)}$ and hypothesis $W_{k,n}$, and $P_{\Lambda}(W_{k,n} | O_k)$ is the posterior probability of each hypothesis according to the model parameters Λ .

By optimizing the LM parameters within the ASR system, MWE training directly targets the reduction of transcription errors, leading to statistically significant improvements in WER compared to conventional cross-entropy training.

Chiu et al. [110] introduced a methodology trains sequence-to-sequence ASR models directly to minimize WER rather than cross-entropy loss. Using reinforcement learning techniques or expected risk minimization, the model optimizes parameters with respect to WER, which better aligns training objectives with evaluation metrics. This improves robustness to recognition errors and better end-to-end ASR performance.

Label Smoothing (LS) An effective remedy for the tendency of sequence-to-sequence ASR models to become over-confident is *labeling smoothing*, in which the one-hot ground-truth distribution is relaxed by assigning a fraction β of the probability mass to the correct token and redistributing the remaining $1 - \beta$ across plausible alternatives. Beyond *uniform* proposed in [111] and *unigram* in [112], smoothing in [29] introduces *neighborhood smoothing*, wherein the residual probability is concentrated on tokens within ± 1 and ± 2 positions in the reference transcript, thereby reflecting the temporal structure of speech. By discouraging the model from collapsing its mass onto a single token, neighborhood smoothing both regularizes learning—yielding nearly a 3% reduction in character error under greedy decoding—and raises the entropy of its output distribution. A small grid search over β settings demonstrates robust performance across a broad range of parameter choices, underscoring the practical value of smooth labeling in ASR decoding.

3 Adaptation

General models often misrecognize specialized terminology and rare words in domain-specific contexts. It makes adaptation techniques essential for optimizing ASR performance, which can be broadly categorized into three main trends: training with domain-specific data, capturing specific phrases or words, and leveraging large language models (LLMs) through prompting schemes. This section explores these schemes, their effectiveness, and the challenges they face in enhancing ASR performance across diverse applications.

3.1 Domain-Specific Training

This approach involves training or fine-tuning ASR models or external LMs with domain-specific paired or text-only data to improve performance in a target domain. A key challenge is the limited availability of domain-specific data, particularly for retraining ASR systems. For external LMs used in the correction tasks, the scarcity of noisy-clean text pairs (generated by ASR) poses a similar issue, prompting innovative solutions to mitigate this constraint.

Mai et al. [114] address the data scarcity challenge by introducing an unsupervised method that employs an encoder-decoder correction network. This approach uses pseudo-labeled data, generated by transcribing audio with inferior ASR models (sub-optimal inputs) and a superior model (optimal targets), to train the correction network without ground truth labels, enhancing domain-specific performance.

Nanayakkara et al. in [67] also focus on the medical domain, evaluating sequence-to-sequence encoder-decoder models for the post-error correction. They highlight the poor performance of pretrained LMs in clinical contexts and demonstrate the need for fine-tuning with domain-specific text-only data. To overcome the lack of ASR transcribed data, they explore fine-tuning tasks like summarization, paraphrasing, and mask-filling, with mask-filling proving most effective due to its similarity with error correction objectives.

Jia et al., in EPIC [115], fine-tune a pretrained language model on classroom-specific synthetic error patterns. This LM is then used to generate multiple correction candidates; a learned similarity score between each candidate and the original hypothesis is finally applied to pick the most contextually plausible correction candidate.

3.2 Capturing Specific Phrases or Words

The methods in this trend prioritize the accurate recognition of a predefined list of phrases or words, often without aiming to enhance overall domain performance. They are particularly valuable in domains with critical terminology, though their effectiveness is limited by the scope of the provided list. However, some approaches introduce selection mechanisms to handle large corpora and improve scalability.

He et al. [18] introduce ED-CEC, a two-branch framework that enhances rare-word recognition by detecting and correcting errors contextually. A BERT-based classifier identifies error positions, followed by AR decoding and augmentation with a domain-specific rare-word lexicon, optimizing computational efficiency by targeting predicted error locations. Zhou et al. [116] propose a human-in-the-loop approach for generating police reports, ensuring accuracy by allowing users to enter or verify specific keywords. This method relies on manual input to capture critical terms, bypassing extensive model retraining. Wang et al. [117] present a contextual spelling correction (CSC) framework with two variants: AR kind of CSC (AR-CSC) and NAR Fast CSC (FCSC). Using a shared text encoder for transcripts and a context encoder for domain-specific phrases, they incorporate ranking mechanisms to select relevant phrases, addressing scalability challenges for large lexicons.

Antonova et al. [118] develop SpellMapper, a NAR spellchecker that uses a pre-built mapping index—created via GIZA++ alignments on TTS-synthesized and ASR-decoded phrase pairs—to correct misspelled phrases. It first selects the correction candidates using character-level n-grams and then decides on replacements using a BERT-based classifier. This enables domain-customized corrections without modifying the acoustic model, though it requires an initial preprocessing step to create the mapping index.

3.3 LLM-Based Methods with Prompting

The main approach in this category leverages the generalizability of LLMs in different domains to refine ASR outputs through various prompting schemes, often without requiring domain-specific training. This approach relies on the broad knowledge embedded in LLMs, adapting them to specific domains via prompting strategies.

Ebadi et al. [119] utilize zero-shot and few-shot prompting with LLMs to enhance transcription accuracy in the medical domain. They also evaluate the impact on named entity recognition tasks, showcasing the flexibility of LLMs in domain-specific refinement without retraining. Adedeji et al. [120] employ Chain-of-Thought (CoT) prompting with LLMs for ASR post-refinement in medical domain, demonstrating superior performance over zero-shot prompting. This approach harnesses structured reasoning to improve correction accuracy across domains.

4 Datasets

To train and evaluate models for ASR error correction, specialized appropriate datasets are crucially required. Approaches for dataset creation range from collecting real-world benchmarks derived from actual ASR system outputs and their corresponding corrected versions [121]. These datasets often reflect errors specific to certain domains, languages, accents, ASR systems, speaker variabilities, or acoustic conditions. Another common method is generating data synthetically using Text-to-Speech (TTS) systems followed by ASR processing to create paired noisy/clean text. Furthermore, pseudo-data is frequently constructed by introducing synthetic errors into large volumes of clean text, often leveraging large language models or observed error patterns to mimic ASR outputs.

This section details the diverse methodologies employed to create and utilize such datasets, which are fundamental for advancing the field. Furthermore, strategies for filtering and refining these datasets to enhance the robustness and performance of the EC models are discussed, ensuring that the training data accurately reflects the complexities and nuances of real-world ASR errors.

4.1 Real-World Benchmark Datasets

Real-world benchmark datasets for ASR are predominantly constructed by leveraging pretrained ASR models or by training custom ASR models on public or private speech datasets to generate ASR ground-truth pairs [47, 92, 122]. Most research in this area utilizes established ASR systems to transcribe speech corpora, creating paired datasets of ASR hypotheses and their corresponding ground-truth transcripts. These datasets are designed to capture authentic ASR errors that arise from diverse real-world conditions, such as background noise, variations among speakers, and acoustic distortions. To foster broader research adoption, some researchers publicly release these generated datasets. Furthermore, many approaches incorporate data augmentation techniques, including the addition of background noise, speed perturbation, and other acoustic modifications, to further introduce realistic ASR errors and enhance dataset diversity, making them suitable for evaluating model performance in practical scenarios.

Table 2: An overview of datasets used for evaluation of ASR refinement models in the literature

Title	Year	Language	Primary Application	No. Samples or Duration	Additional Information	Cite
WSJ	1993	English	Speech Recognition	80 hours	read speech from WSJ, news domain	[137]
ATIS	1993	English	Speech Recognition	34 hours	spoken language, travel domain	[138]
LVCSSR	1996	English	Speech Recognition	40 calls	conversational speech, telephone calls	[139]
Switchboard	1997	English	Speech Recognition	260 hours	conversational speech, telephone calls	[140]
Fisher	2004-2005	English	Speech Recognition	1,960 hours	conversational speech, telephone calls, manual transcript	[141]
AMI	2005	English	Speech Recognition	100 hours	meetings, multi-speaker, manual transcript	[142]
CHiME-1	2011	English	Speech Recognition	19.7 hours	read speech (plus noise and RIR) from WSJ, news domain, manual transcript	[143]
TED-LIUM 1	2012	English	Speech Recognition	118 hours	conference speech, auto alignment	[144]
CHiME-2	2013	English	Speech Recognition	324 hours	news domain, manual transcript	[145]
TED-LIUM 2	2014	English	Speech Recognition	207 hours	conference speech, manual and auto alignment transcript	[146]
LibriSpeech	2015	English	Speech Recognition	1,000 hours	read speech from audiobooks	[125]
CHiME-3	2015	English	Speech Recognition	13.3 hours	noisy locations (e.g., bus, street), news domain, manual transcript	[147]
CHiME-4	2016	English	Speech Recognition	13.3 hours	news domain, same as CHiME-3 but multi-channel, manual transcript	[147]
SWC	2016	English	Speech Recognition	395 hours	volunteer readers, encyclopedia domain	[148]
LRS2	2017	English	Speech Recognition	224 hours	audio-visual speech, news and shows domain	[149]
CHiME-5	2018	English	Speech Recognition	49 hours	conversational dinner party, manual transcript	[150]
TED-LIUM 3	2018	English	Speech Recognition	452 hours	conference speech, manual and auto alignment transcript	[151]
SWGSpeech	2021	English	Speech Recognition	5000 hours	finance domain, manual transcript	[152]
PriMock 57	2022	English	Speech Recognition	8.5 hours	primary-care telemedicine domain, manual transcript	[153]
HyParadise (HP)	2023	English	Text Refinement	334,000 samples	general (mixed) domain	[154]
SIGHAN13	2013	Chinese	Text Refinement	1700 samples	language learning domain (TOCFL), L2 learners	[155]
SIGHAN14	2014	Chinese	Text Refinement	4497 samples	language learning domain (TOCFL), L2 learners	[156]
SIGHAN15	2015	Chinese	Text Refinement	3439 samples	language learning domain (TOCFL), L2 learners	[157]
AISHELL-1	2017	Chinese	Speech Recognition	178 hours	read speech, several domains (e.g., finance, technology, sports, news)	[158]
AISHELL-2	2018	Chinese	Speech Recognition	1,000 hours	read speech, several domains (e.g., finance, technology, sports, news)	[159]
Wang211K	2018	Chinese	Text Refinement	271,299 samples	general domain (news-inclusive), symbolic (OCR) and phonetic (ASR) errors	[160]
Aidatatang	2019	Chinese	Speech Recognition	200 hours	read speech, mostly mobile records, manual transcript	[161]
AISHELL-3	2020	Chinese	Speech Recognition	85 hours	general domain, multi-speaker	[162]
AISHELL-4	2021	Chinese	Speech Recognition	120 hours	conference/meeting speech, multi-channel	[163]
LEMON	2023	Chinese	Text Refinement	22,000 samples	from human writings, manual transcript, multi-domain (medical-inclusive)	[164]
ChFT	2024	Chinese	Text Refinement	562,048 samples	news domain	[165]
ASR-EC	2024	Chinese	Text Refinement	1,000,000 samples	general (mixed) domain	[24]
CSCD-NS	2024	Chinese	Text Refinement	40,000 samples	social media posts domain, native speakers	[166]
AISHELL-5	2025	Chinese	Speech Recognition	100 hours	in-car recordings, multi-channel, noise included	[167]
CSJ	2004	Japanese	Speech Recognition	661 hours	formal speech, academic domain	[168]
ALFFA	2005	Amharic	Speech Recognition	20 hours	newspaper/magazine domain, manual transcript	[169]
ALFFA	2012	Swahili	Speech Recognition	11.75 hours	newspaper/magazine domain, manual transcript	[170]
ALFFA	2016	Wolof	Speech Recognition	16.8 hours	newspaper/magazine domain, manual transcript	[171]
SWC	2016	German	Speech Recognition	386 hours	volunteer readers, encyclopedia domain	[148]
SWC	2016	Deutsch	Speech Recognition	224 hours	volunteer readers, encyclopedia domain	[148]
Common Voice	2020	129 languages	Speech Recognition	33,000 hours	general domain	[172]
SPREDS-UI	2023	21 languages	Speech Recognition	tens of thousand (not stated)	not clearly stated	[173]
FLEURS	2023	102 languages	Speech Recognition	12 hours per language	encyclopedia	[174]

Two significant examples of datasets created using industry-grade ASRs to generate ASR-ground truth pairs are HyParadise (HP) and ASR-EC. Chen et al. [121] developed the HP dataset by generating N-best hypotheses (top-5 from a beam search with a beam size of 60) from two state-of-the-art ASR models, WavLM [123] and Whisper [124]. This process has been applied to numerous popular ASR datasets, including LibriSpeech [125], CHiME-4 [126], WSJ [127], SwitchBoard [128], Common Voice [129], Tedlium-3 [130], LRS2 [131], ATIS [132], and CORAAL [133], yielding over 334,000 pairs. In a similar vein, the ASR-EC benchmark [134], specifically designed for Chinese ASR error correction, was constructed by collecting erroneous transcriptions from industry-grade ASR systems and pairing them with manually verified ground-truth transcripts. ASR-EC utilizes data from diverse Chinese speech corpora, such as AISHELL-1 [135] and WenetSpeech [136], to cover a wide spectrum of ASR errors. An extensive list of ASR employed by different researches for ASR refinement is presented in Table 4.

4.2 TTS-ASR Synthesis Methods

TTS-ASR synthesis methods offer a versatile and scalable framework for generating synthetic datasets to enhance ASR error correction, particularly excelling in context-specific scenarios and full-text applications. This approach begins with converting clean text into speech using TTS systems, which can be fine-tuned to incorporate diverse variables such as speaker identity, accent, and background noise. The resulting audio is then transcribed back into text by an ASR model, yielding noisy hypotheses that replicate real-world transcription errors. This flexibility enables the creation of tailored datasets for training robust error correction models, accommodating challenges like multi-speaker environments or domain-specific nuances. By systematically manipulating TTS parameters, researchers can simulate a broad spectrum of error conditions, making TTS-ASR synthesis a foundation for developing advanced correction systems that address both localized and generalized ASR inaccuracies.

Numerous studies have harnessed TTS-ASR synthesis, leveraging multi-speaker TTS systems and additional augmentations to tackle specialized error correction tasks. For example, [19] employed a multi-speaker, multi-locale TTS to synthesize context phrases (e.g., names), pairing them with ASR-generated hypotheses to train a contextual spelling correction model, embedding these pairs into sentence patterns for efficiency. Similarly, [175] developed the

ChFT dataset for full-text Chinese error correction, using a TTS-ASR pipeline with ChatTTS and a cascaded ASR system, incorporating voice activity detection and punctuation restoration to span multiple domains. [176] advanced this further by integrating multi-speaker TTS (e.g., YourTTS) with spectrogram-level augmentations like frequency masking, generating large-scale synthetic data for their denoising LM, achieving top performance on LibriSpeech. Meanwhile, [177] targeted context-specific scenarios by synthesizing speech with placeholders (e.g., names) and augmenting data via word replacement, addressing data sparsity. These efforts highlight how TTS-ASR synthesis, enhanced by multi-speaker systems and creative augmentations, drives innovation in error correction across diverse applications.

4.3 Pseudo Data Generation

Given the scarcity of large-scale paired ASR error and correction datasets, pseudo data generation has become essential for training robust error correction models. Previous works such as [83, 178, 179] have demonstrated the effectiveness of synthetic data in improving ASR correction performance. Early approaches like [180] create benchmark datasets from public sources, utilizing the Amazon Review Dataset [181] and Yelp Open Dataset [182] with 100,000 unique entities each. They introduce realistic ASR errors through an n-gram confusion matrix learned from commercial dialog systems, carefully partitioning error patterns between training and testing to prevent simple memorization.

More sophisticated approaches leverage NLMs for pseudo-data generation. [183] employs a BERT model to automatically generate error tokens, incorporating homophone substitutions to simulate common ASR errors. Their approach involves training a 12-layer BERT generator with masked language modeling, where 20% of tokens undergo noise injection following a specific distribution: 10% unchanged, 40% masked, 40% replaced with homophones, and 10% randomly substituted. During generation, this model produces three types of ASR errors—deletion, insertion, and substitution—by strategically masking tokens and using BERT’s predictions as pseudo-errors, creating more realistic training data compared to random perturbation methods.

Recent work has explored the use of LLMs to generate synthetic ASR correction data when labeled resources are limited. In [115], error patterns are first extracted from real ASR outputs and then are used to guide LLMs in generating diverse, contextually grounded sentences that mimic realistic ASR errors. This strategy enables scalable data augmentation and improves the correction robustness, particularly in domain-specific and low-resource scenarios like classroom dialogue transcription.

4.4 Data Filtering Strategies for EC Training

Automatically pairing ASR hypotheses with reference transcripts is a common method for generating EC datasets, but such data often includes noisy or non-inferable examples. Udagawa et al. [93] propose a conservative data filtering approach based on two criteria: (1) the correction should improve linguistic acceptability, and (2) it must be inferable from the ASR hypothesis and its phonemes. Pairs that fail either criterion are retained without modification to train the model to avoid unnecessary corrections. This method significantly reduces overcorrection, particularly in out-of-domain scenarios, and outperforms traditional filtering approaches based on the edit distance. The strategy emphasizes dataset quality over quantity and has proven effective across different LLMs and test domains.

5 Metrics

Refinement techniques are essential for enhancing the accuracy and usability of ASR systems, requiring evaluation metrics that are sufficiently sensitive to key errors. These metrics can be systematically categorized into four classes: error rate metrics, language quality metrics, entity metrics, and qualitative review frameworks. This categorization, detailed below, facilitates the selection of appropriate tools to evaluate ASR refinement methods across various domains of performance, ensuring both precision and applicability in diverse contexts. An overview of this section is provided in Figure 4, and Table 3 presents illustrative examples that underscore the necessity of carefully examining evaluation metrics.

5.1 Error Rate Metrics

This type of metrics is on quantifying transcription accuracy by measuring discrepancies between refined ASR outputs and reference texts. These metrics are foundational in assessing the correctness of transcriptions at both word and character levels, often serving as benchmarks for system improvement.

- **WER & WERR:** WER is the percentage of substitutions (S), deletions (D), and insertions (I) required to align hypothesis and reference, normalized by the reference length N :

$$\text{WER} = \frac{S + D + I}{N} \times 100.$$

These error counts are obtained via a standard Levenshtein alignment. WER Reduction (WERR) quantifies relative improvement:

$$\text{WERR} = \frac{\text{WER}_{\text{before}} - \text{WER}_{\text{after}}}{\text{WER}_{\text{before}}} \times 100.$$

Domain-specific variants simply reapply the same formula to a targeted unit set. For example, Medical Concept WER (MC-WER), utilized in [87] treats each medical concept as one atomic “word”, counting S_m, D_m, I_m over M reference concepts:

$$\text{MC-WER} = \frac{S_m + D_m + I_m}{M} \times 100,$$

thereby customizing WER to medical domain without altering its core definition.

- **CER & CERR:** Character Error Rate (CER) applies the same formula at the character level, capturing spelling and morphological mistakes. Character Error Rate Reduction (CERR) is defined analogously to WERR, expressing the percent reduction in CER.
- **Stop Word Filtered WER (swf-WER):** Computes WER after removing common stop words (e.g., “the,” “and”) from both hypothesis and reference, focusing the metric on semantically informative terms [62].
- **Corrected Percentage WER (cpWER):** Measures the fraction of originally erroneous words that are corrected by a refinement method, highlighting its corrective impact [184].

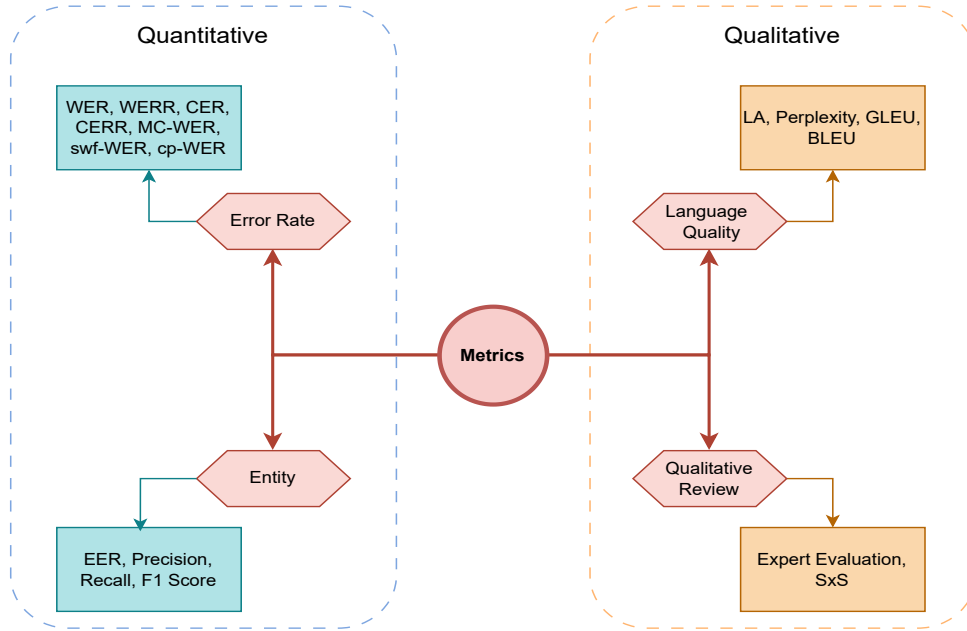


Figure 4: Overview of metrics used to evaluate ASR refinement.

5.2 Entity Metrics

These metrics target the accuracy of identifying specific entities (e.g., names, dates, or locations) within refined transcriptions, critical for applications requiring structured information extraction.

- **Entity Error Rate (EER):** Quantifies errors in entity recognition, calculated as the proportion of incorrectly identified entities relative to the total.

- **Precision:** Measures the fraction of proposed corrections that are true positives. In the ASR refinement context, this indicates how often the model’s changes correctly introduce a missing user phrase, rather than hallucinating new content:

$$\text{Precision} = \frac{\text{better}}{\text{better} + \text{fp}},$$

where *better* is the number of previously missing phrases correctly introduced by the refinement and *fp* is the number of phrases erroneously added [118].

- **Recall:** Measures the fraction of actual missing phrases that the model successfully recovers. This reflects the model’s ability to detect and correct errors in the original ASR output:

$$\text{Recall} = \frac{\text{better}}{\text{better} + \text{missed}},$$

where *missed* is the number of originally missing phrases that remain uncorrected and *better* is as mentioned above.

- **F1 Score:** The harmonic mean of precision and recall, offering a balanced evaluation of entity recognition performance:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Table 3: Illustrating gaps and the importance of each metric category for ASR refinement evaluation.

Illustrative Examples	Example 1	Example 2	Example 3	Example 4
Ground Truth	the cat sat on the mat	Dr. Smith prescribed antibiotics like amoxicillin for the patient	The Battle of Hastings was fought in 1066	The patient has a history of diabetes
ASR Transcript	a cut start in a math	Doctor described anti bio things like amoxilin for the patience	The Battle of Hasting was fought in 1966	The patient has a history of diabetics
Refiner1	a cat sat in a mat	Dr. Smith described antibiotics like amoxicillin for the patient	The Battle of Hastings was fought in 1966	The patient has a history of diabetes
Refiner2	the cut start on the math	Doctor Smith prescribed anti bio think like amoxilin for the patient	The Battle of Hasting was in 1066	The patient has a history of diabetes and hypertension
Category with Gap	Error Rate Metrics	Entity Metrics	Language Quality	Qualitative Review
Explanation of Gap	Both refiners have the same WER (e.g., 4 errors out of 6 words), but Refiner1 corrects more meaningful content words ("cat," "sat," "mat"). WER treats all errors equally, missing semantic importance.	Refiner1 has better EER (correctly identifies "Dr. Smith," "amoxicillin"), but "described" is contextually wrong. Entity Metrics focus on entities, missing the critical verb error.	Both refiners produce fluent outputs (similar BLEU scores), but Refiner2 corrects the critical year "1066." Language Quality Metrics prioritize fluency over factual accuracy.	Refiner2 adds "and hypertension" (hallucination), which might be rated plausible by evaluators without ground truth. Qualitative Review can be subjective and miss factual errors.
Complementary Metrics	Entity Metrics (for key terms like "cat," "mat"), swf-WER (focuses on content words)	Language Quality Metrics (e.g., BLEU for coherence), Qualitative Review (for contextual accuracy)	Error Rate Metrics (e.g., WER detects year error), Entity Metrics (for "1066")	Error Rate Metrics (e.g., WER detects insertion), Entity Metrics (Precision flags hallucination)

5.3 Language Quality Metrics

These metrics assess the linguistic fluency, naturalness, and coherence of refined ASR transcriptions, moving beyond mere accuracy to evaluate how well outputs resemble human-like language. These are often employed to gauge usability in real-world applications.

- **Language Acceptability (LA):** A subjective metric, typically evaluated by LM, assessing the fluency and naturalness of transcriptions [93].
- **Bilingual Evaluation Understudy (BLEU):** Originally from machine translation, BLEU measures n-gram overlap between ASR output and reference texts, providing a quantitative measure of fluency and adequacy [75, 185–187].

- GLEU: It addresses BLEU’s instability on single sentences by extracting up to 4-grams, computing both precision and recall, and taking their minimum. The resulting symmetric, $[0, 1]$ -bounded score requires no smoothing and delivers stable per-sentence rewards for reinforcement-learning in sequence models [75].
- Perplexity: Derived from language modeling, this metric evaluates how well a model predicts a sample, with lower perplexity indicating better alignment with expected language patterns [10].

5.4 Qualitative Review Frameworks

These frameworks address unique evaluation needs, incorporating human judgment to capture aspects not covered by objective categories mentioned earlier.

- Expert Evaluation: Involves domain experts reviewing transcriptions for quality, leveraging human insight to assess context-specific nuances [188].
- Side-by-Side (SxS): A comparative method where human evaluators rate two transcriptions side by side, providing a direct quality comparison [189].

6 Research Gap and Future Work

Despite significant advancements in ASR refinement techniques, several critical research gaps are remaining that warrant further investigation. A key limitation in the current literature is the predominant use of text-only data in refinement methods. Many techniques rely on general LMs to correct ASR outputs, but this approach often falls short because the text output from ASR systems does not always conform to the linguistic structures encoded in these models. Human speech, particularly in everyday informal conversations, frequently exhibits varied and unconventional patterns that differ from the formal structures on which the general LMs are trained on. As a result, applying such models can introduce overcorrection that inadvertently increases the error rate, especially in scenarios where the WER is quite high and errors stem from insertions and deletions.

While some methods employ detection modules or confidence estimation to prevent overcorrection, these components are typically built on text-only data as well, inheriting the same limitations. This issue is particularly pronounced in techniques leveraging LLMs, which, despite improvements in training or fine-tuning for specific domains, struggle in high-error-rate contexts and out-of-domain evaluations. Consequently, there is a clear need for approaches that can better guide the correction process, particularly in scenarios with high WER, frequent insertions and deletions, and out-of-domain evaluations. This is because there is no clear assessment of error types before and after applying refinement methods; only a small number of studies have evaluated the post-refinement error types. Therefore, more research is needed to classify and evaluate errors across different domains, including clean versus noisy environments and general versus specific contexts. Additionally, examining how different modalities—text, phonetics, and speech—affect refinement techniques for each error type could shape the future work and help users select the most appropriate models for their needs and computational resources.

Another underexplored area is the computational efficiency of ASR refinement techniques. Although NAR models and detection modules have been introduced to accelerate the correction process, only a limited number of studies have prioritized this critical factor. The field would benefit from a detailed evaluation of methods that strike a balance between effectiveness and implementation costs. Such an assessment would ensure that refinement techniques are not only accurate but also feasible for practical, real-world deployment, where computational resources and processing speed are often constrained.

Furthermore, the use of contextual features has largely been restricted to text-based methods, with only a handful of recent studies exploring contextual features derived from speech data. Investigating fine-grained contextual speech features, beyond coarse representations, presents a promising avenue for enhancing the refinement process. By leveraging richer acoustic and prosodic cues from speech, future techniques could overcome some of the limitations inherent in text-only approaches, potentially leading to more robust and accurate corrections.

A significant challenge in the field is the difficulty in comparing existing methods without re-implementation, which is often impractical for researchers and end-users. This stems from the lack of standardized evaluation practices, as different studies employ diverse datasets and metrics. Such inconsistencies obscure the true effectiveness of proposed techniques and hinder progress by preventing fair and direct comparisons. To address this, there is a crucial need for a unified leaderboard or benchmarking framework that standardizes the evaluation of ASR refinement methods across common datasets and metrics. This would enable researchers to systematically assess and compare techniques, fostering the development of more reliable and efficient solutions.

In summary, future research should prioritize the development of cross-modal refinement techniques capable of addressing the complexities of informal human speech, particularly in high-error-rate scenarios. Efforts should also focus on enhancing the computational efficiency of these methods and exploring the potential of contextual speech features. Finally, establishing a standardized evaluation framework will be crucial for ensuring consistency and facilitating meaningful comparisons across the field. By tackling these gaps, the ASR refinement community can move toward more effective, practical, and widely applicable solutions.

7 Conclusion

This survey provides a comprehensive overview of non-intrusive ASR refinement techniques, categorizing them into five classes: fusion, rescoring, correction, distillation, and training adjustment. Additionally, we have explored domain-specific adaptation techniques and provided a detailed review of datasets, creation schemes, and evaluation metrics used in ASR refinement research. Our work addresses key gaps in the literature by offering a structured analysis of these methods, their strengths, limitations, and applicability in various environments. This survey also provides a baseline for future research by reviewing datasets, their creation processes, and filtering schemes, as well as an extensive categorization of evaluation metrics. This enables researchers to select appropriate metrics for consistent comparative studies in the field.

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